Example 6.21

The time to failure in thousands of hours of an important piece of electronic equipment used in a manufactured DVD player has the density function.

$$f(x)=\begin{cases}3e^{-3x}, & x>0\\ 0, & otherwise\end{cases}$$

Find the expected life of the piece of equipment.

Solution:

We know that,

$$E(X)=\int_{-\infty}^{\infty} x\,f(x)\,dx$$

$$=\int_{0}^{\infty} x\,3e^{-3x}dx$$

$$=3\int_{0}^{\infty} x\,e^{-3x}dx$$

$$=3\left\{\left[x\frac{e^{-3x}}{-3}\right]_0^{\infty}-\int_0^{\infty}\left(\frac{e^{-3x}}{-3}\right)dx\right\}\quad\left(\because\int u\,dv=uv-\int v\,du\right)$$

$$=\int_{0}^{\infty} e^{-3x}dx$$

$$=\frac{1}{3}$$

Therefore, the expected life of the piece of equipment is $\frac{1}{3}$ hrs (in thousands).

Example 6.22

A commuter train arrives punctually at a station every 25 minutes. Each morning, a commuter leaves his house and casually walks to the train station. Let X denote the amount of time, in minutes, that commuter waits for the train from the time he reaches the train station. It is known that the probability density function of X is

$$f(x)=\begin{cases}\frac{1}{25}, & for\ 0<x<25\\ 0, & otherwise.\end{cases}$$

Obtain and interpret the expected value of the random variable X.

Solution:

Expected value of the random variable is

$$E(X)=\int_{-\infty}^{\infty} x\,f(x)\,dx$$

$$=\int_{0}^{25} x\frac{1}{25}dx$$

$$=\frac{1}{25}\int_{0}^{25} x\,dx$$

$$=\frac{1}{25}\left[\frac{x^2}{2}\right]_0^{25}$$

$$=12.5$$

Therefore, the expected waiting time of the commuter is 12.5 minutes.

Example 6.23

Suppose the life in hours of a radio tube has the probability density function

$$f(x)=\begin{cases}e^{-\frac{x}{100}}, & when\ x\geq 100\\ 0, & when\ x<100\end{cases}$$

Find the mean of the life of a radio tube.

Solution:

We know that, the expected random variable

$$E(X)=\int_{-\infty}^{\infty} x\,f(x)\,dx$$

$$=\int_{100}^{\infty} x\,e^{-\frac{x}{100}}dx$$

$$=\left\{\left[x\left(\frac{e^{-\frac{x}{100}}}{-\frac{1}{100}}\right)\right]_{100}^{\infty}-\int_{100}^{\infty}\left(\frac{e^{-\frac{x}{100}}}{-\frac{1}{100}}\right)dx\right\}$$

$$\left(\because\int u\,dv=uv-\int v\,du\right)$$

$$=\left[(10000)\left(e^{-1}\right)+(10000)\left(e^{-1}\right)\right]$$

$$=\left[(10000)(0.3679)+(10000)(0.3679)\right]$$

$$=7358\ hours$$

Therefore, the mean life of a radio tube is 7,358 hours.

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